SYNTHETIC COVID-19 TRIAGE RISK PREDICTION

Machine Learning with Radiomics-Style, Clinical and Laboratory Features

Project Report

Inspired by *CT-based Patient Triage of COVID-19: Radiomics Prediction of ICU Admission, Mechanical Ventilation, and Death of Patients* by Xianghao Zhan and Yiheng Li.

Data limitation

This project uses a synthetically generated dataset. It does not use the original patient-level cohort or real CT images from the reference paper. Radiomics-style variables are simulated numeric features. Results are for educational demonstration only and are not clinically validated.

Student: Renetta Isabel Nathan
SRN: PES1UG24AM221
Course / subject: [Add course name]
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Abstract

This project implements an educational machine-learning workflow inspired by research on COVID-19 patient triage using radiomics, clinical variables, and laboratory measurements. It predicts three binary outcomes: ICU admission, mechanical ventilation, and death. Since the original patient-level data were not used, a synthetic dataset of 2,400 rows and 155 columns was generated. The workflow includes validation, exploratory data analysis, feature-group construction, preprocessing, SMOTEENN class-imbalance handling on training data, LASSO-based feature selection, model training, evaluation, and permutation-importance analysis. Five classifiers were compared across three feature groups and three outcomes, yielding 45 experiments. The top reported ROC-AUC values were 0.721 for ICU admission, 0.737 for mechanical ventilation, and 0.730 for death. These results describe the synthetic experiment only and are not evidence of clinical performance.

1. Introduction

Severe COVID-19 outcomes can include intensive care admission, mechanical ventilation, and death. The reference paper explores combining CT-derived radiomics with clinical and laboratory information for patient triage. This project adapts the broad modelling workflow as a coursework exercise using synthetic tabular data rather than the original clinical cohort or actual CT images.

2. Objectives

The objectives are to validate a dataset; explore target distributions, missingness, and correlations; compare three feature groups; preprocess variables; address training-set class imbalance; apply LASSO feature selection; train five classifiers; compare evaluation metrics; and inspect feature importance.

3. Dataset

The source dataset used in the workflow has 2,400 rows and 155 columns. It contains an identifier, demographic and symptom/comorbidity variables, laboratory measurements, radiomics-style variables, and three targets: ICU, mechanical_ventilation, and death. The project validation screenshots confirm 704 ICU-positive records (29.33%), 670 mechanical-ventilation-positive records (27.92%), and 391 death-positive records (16.29%). These proportions are generated by the synthetic data process and should not be interpreted as real-world prevalence.

A feature-count cell in one screenshot reports 132 radiomics features and 167 total columns, which conflicts with the 155-column dataset preview and the downstream feature-group counts. The final report therefore uses the verified dataset shape and processed matrix dimensions, and flags the count discrepancy for correction in the notebook.

4. Methodology

4.1 Exploratory data analysis

EDA examined outcome distributions, age distribution and age by ICU status, radiomics feature correlations, clinical/laboratory correlations, target correlations, and missing-data patterns. The largest missingness shown was D-dimer at 5.75%, followed by LDH at 4.71%, CRP at 3.75%, creatinine at 3.58%, and albumin at 3.08%. The notebook reported 127 columns with at least one missing value.

The synthetic age-by-ICU boxplot showed a slightly higher median age in the ICU group, but substantial overlap between groups. The target-correlation heatmap showed positive correlations of approximately 0.31 between ICU and mechanical ventilation, 0.22 between ICU and death, and 0.25 between mechanical ventilation and death. These are associations in the generated dataset, not causal or clinical findings.

4.2 Feature groups and preprocessing

The Radiom feature group contains radiomics-style features; RadioClin adds clinical variables; and RadioClinLab adds laboratory variables. The processed matrices had 1,680 training rows and 720 test rows. Radiom used 120 columns, RadioClin used 147, and RadioClinLab used 164 after encoding. A preprocessing output confirmed no missing values remained in the processed RadioClinLab training matrix.

4.3 Class imbalance and LASSO selection

SMOTEENN was applied to training data for each feature-group/target combination. LASSO-based selection used L1-regularized Logistic Regression. Selected features were: Radiom—113 for ICU, 115 for mechanical ventilation, and 109 for death; RadioClin—131, 129, and 128 respectively; RadioClinLab—148, 143, and 145 respectively. Test rows were retained at 720 during feature selection.

4.4 Model training

Five classifiers were trained: Logistic Regression, Random Forest, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and LightGBM. With three feature groups and three targets, this produced 45 model evaluations.

4.5 Evaluation metrics

The workflow reports accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC (Average Precision). ROC curves, precision-recall curves, and confusion matrices were generated to inspect the RadioClinLab ICU Logistic Regression model and compare classifiers.

5. Results

Target	Feature group	Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
ICU	RadioClinLab	Logistic Regression	0.321	0.301	1.000	0.463	0.721	0.509
Mechanical ventilation	RadioClinLab	Logistic Regression	0.343	0.308	0.967	0.467	0.737	0.552
Death	RadioClinLab	Random Forest	0.200	0.172	1.000	0.294	0.730	0.317

5.1 ICU evaluation

For RadioClinLab Logistic Regression, the ROC-AUC was 0.721 and Average Precision was 0.509. The confusion matrix contained 20 true negatives, 489 false positives, 0 false negatives, and 211 true positives. This corresponds to 100% recall but low precision (0.301) and accuracy (0.321) at the selected classification threshold. The model flags nearly all positive ICU cases, but also incorrectly flags many negative cases.

5.2 Model comparison

Among the reported top rows, mechanical-ventilation prediction had the highest ROC-AUC (0.737), followed by death (0.730) and ICU admission (0.721). RadioClinLab appeared among the top feature groups for each outcome in the supplied comparison. Model selection based on test-set metrics can be optimistic; final selection should be performed on validation data and the test set reserved for final assessment.

5.3 Permutation importance

For the RadioClinLab ICU model, the largest mean permutation importance values were at selected feature positions 119 (0.005158), 128 (0.004551), 114 (0.004203), 92 (0.003739), and 93 (0.003724). These values represent mean ROC-AUC decrease after shuffling each selected feature. The supplied output identifies selected-feature positions, not mapped feature names, so no clinical names are assigned to them in this report. Importance is model-dependent and may be unstable when features are correlated.

6. Limitations and ethical considerations

This is a synthetic-data experiment. It does not use the original patient-level cohort, actual CT images, or segmentation masks. Radiomics-style variables are simulated. The metrics depend on the synthetic generator and data split, and cannot establish real-world generalization or clinical usefulness. The project must not be used for diagnosis, triage, or treatment decisions. Correlation and feature-importance results do not establish causation or medical significance.

7. Conclusion

The project demonstrates an end-to-end educational machine-learning workflow: validation, EDA, feature-group preparation, preprocessing, training-set class-imbalance handling, LASSO feature selection, training of five classifiers, evaluation, and permutation-importance analysis. The highest reported ROC-AUC among the supplied top models was 0.737 for mechanical-ventilation prediction. The primary achievement is implementing and examining the workflow on synthetic data, not clinical prediction or replication of the reference paper.

8. Future work

Future work should resolve the feature-count discrepancy in the validation notebook, confirm that model comparison plots and CSV results use the same saved results, choose models using validation data, review classification thresholds and calibration, and map selected-feature positions to their actual feature names before interpreting them. Record package versions and random seeds for reproducibility.

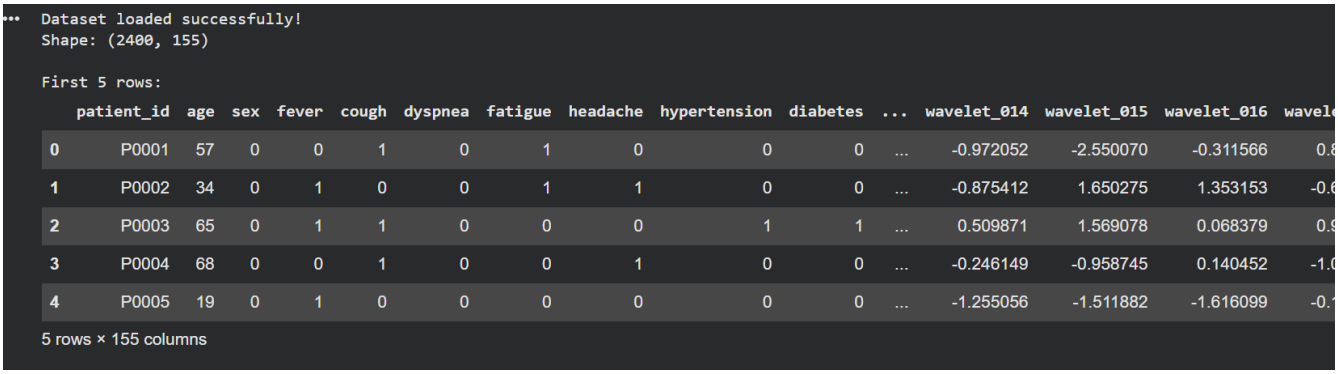
9. Reference

Zhan, X. and Li, Y. “CT-based Patient Triage of COVID-19: Radiomics Prediction of ICU Admission, Mechanical Ventilation, and Death of Patients.” This project is inspired by the broad modelling approach and does not reproduce the original patient cohort.

Appendix A. Notebook Evidence

The following figures are screenshots supplied from the project notebooks. Captions describe the output shown; the screenshots are included as implementation evidence.

Dataset preview



```
Dataset loaded successfully!
Shape: (2400, 155)

First 5 rows:
```

	patient_id	age	sex	fever	cough	dyspnea	fatigue	headache	hypertension	diabetes	...	wavelet_014	wavelet_015	wavelet_016	wavele
0	P0001	57	0	0	1	0	1	0	0	0	...	-0.972052	-2.550070	-0.311566	0.8
1	P0002	34	0	1	0	0	1	1	0	0	...	-0.875412	1.650275	1.353153	-0.6
2	P0003	65	0	1	1	0	0	0	1	1	...	0.509871	1.569078	0.068379	0.9
3	P0004	68	0	0	1	0	0	1	0	0	...	-0.246149	-0.958745	0.140452	-1.0
4	P0005	19	0	1	0	0	0	0	0	0	...	-1.255056	-1.511882	-1.616099	-0.1

5 rows × 155 columns

Dataset preview

Missing-value summary

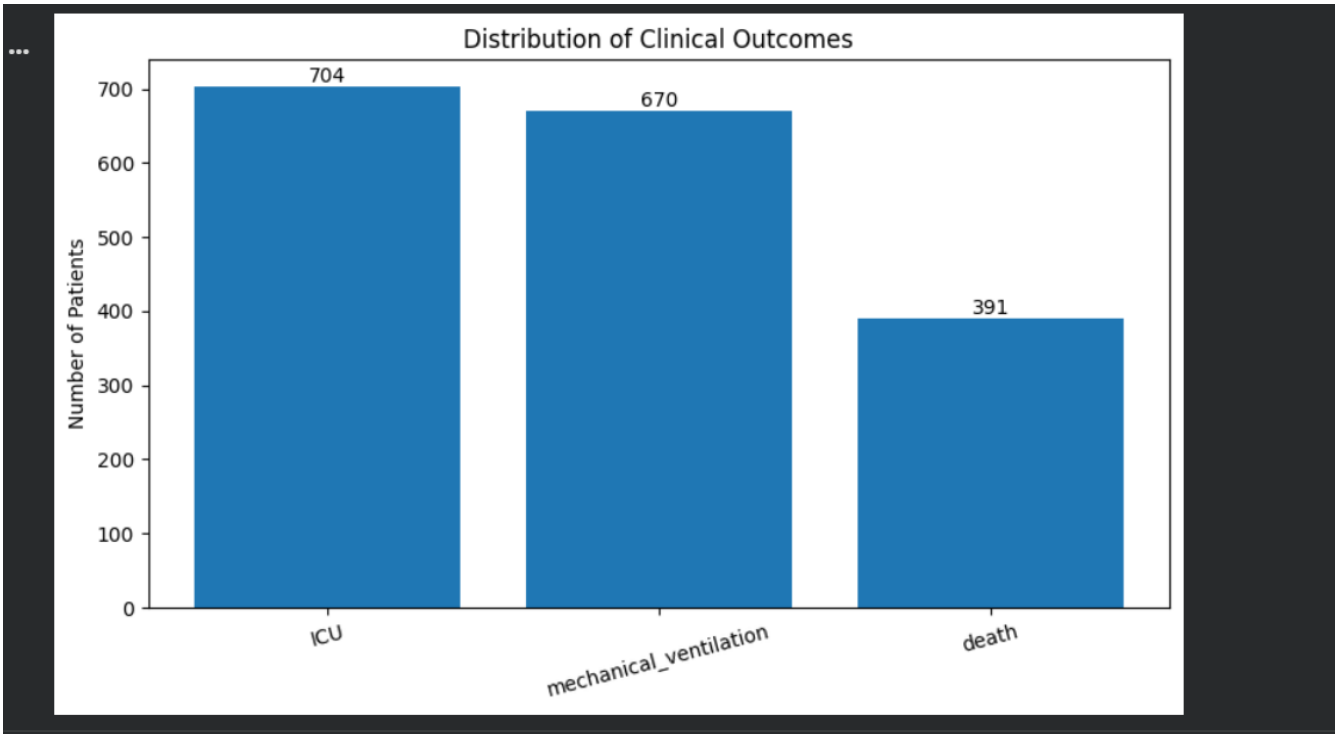
Columns containing missing values:

	Missing_Count	Missing_Percentage
d_dimer	138	5.750000
ldh	113	4.708333
crp	90	3.750000
creatinine	86	3.583333
albumin	74	3.083333
...	...	...
glrlm_007	15	0.625000
wavelet_003	15	0.625000
glrlm_003	14	0.583333
glrlm_015	14	0.583333
wavelet_014	14	0.583333

127 rows × 2 columns

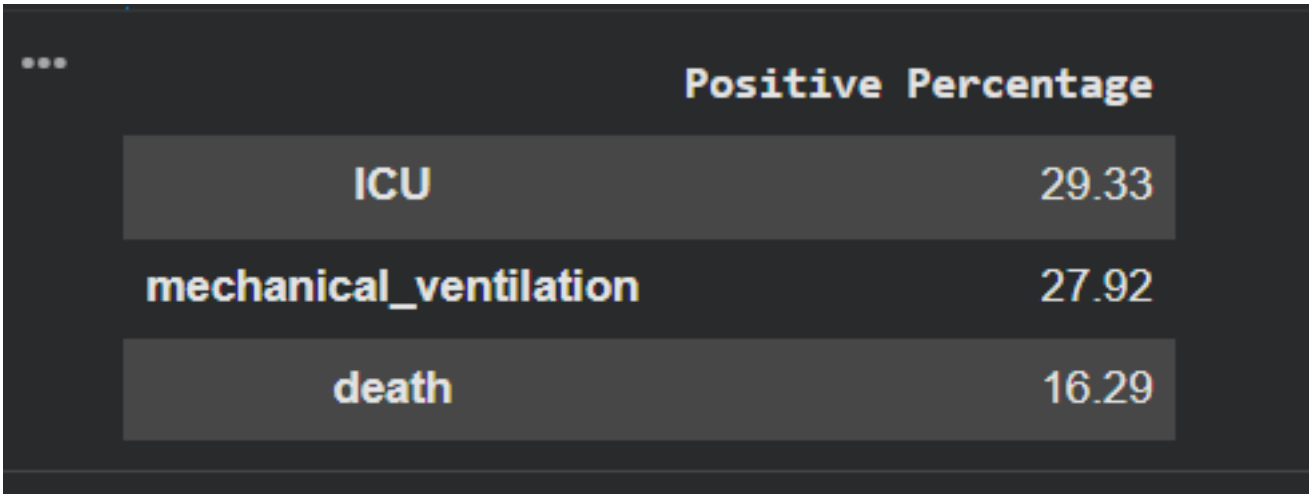
Missing-value summary

Outcome distribution



Outcome distribution

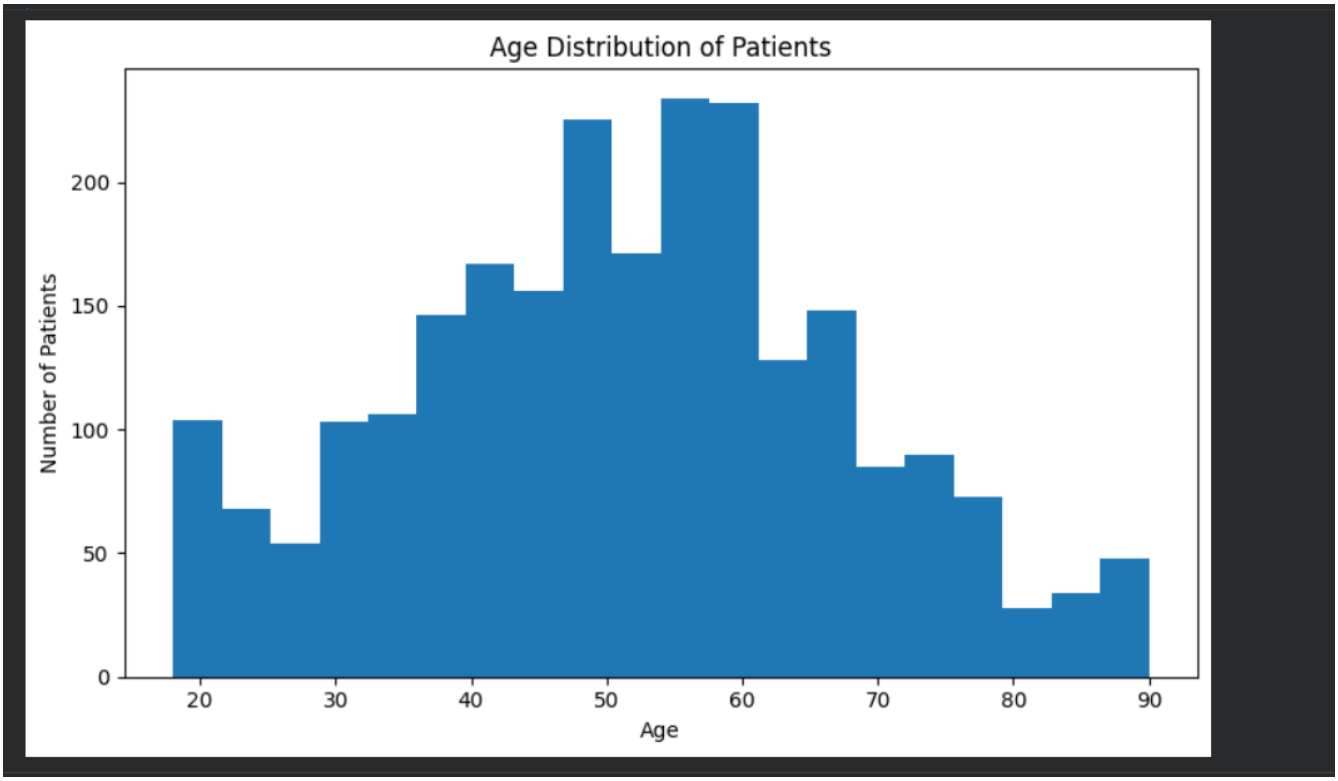
Age distribution



	Positive Percentage
ICU	29.33
mechanical_ventilation	27.92
death	16.29

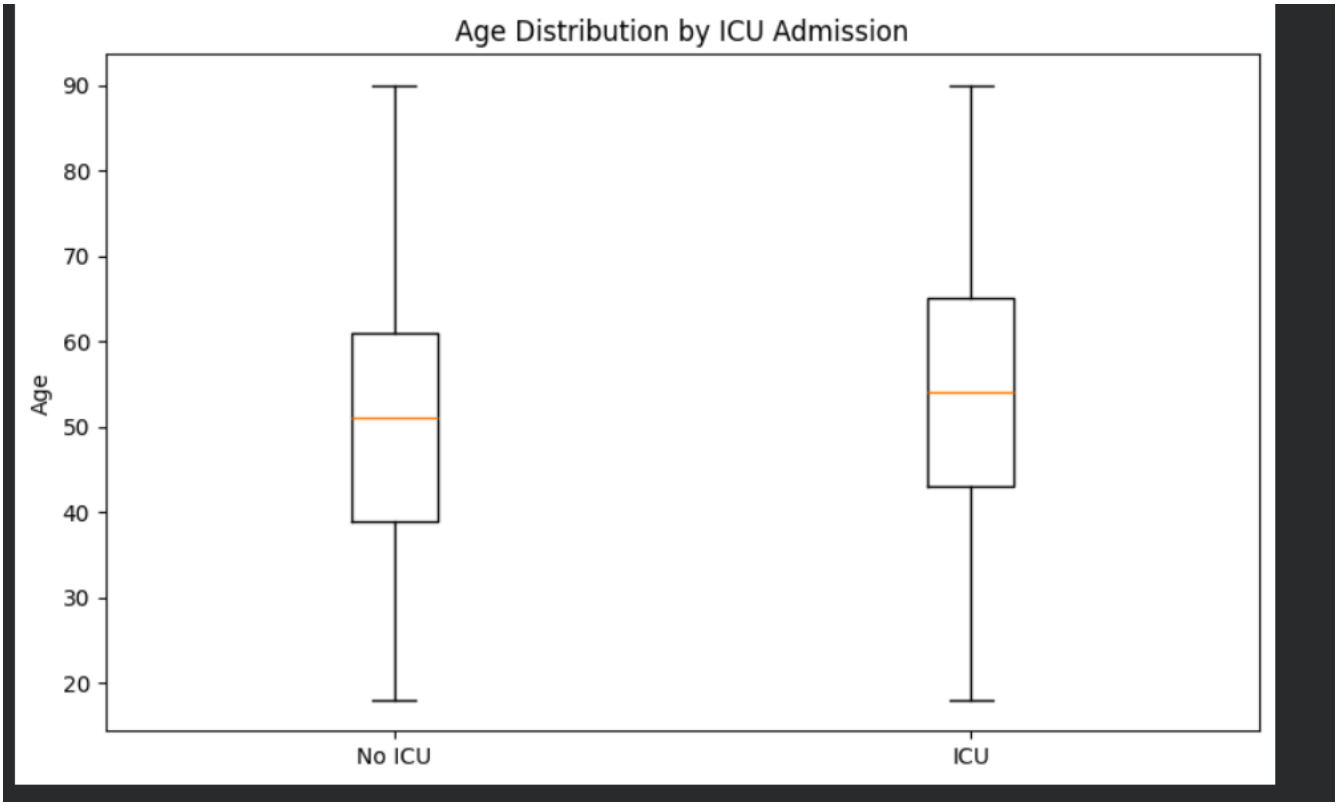
Age distribution

Age by ICU admission



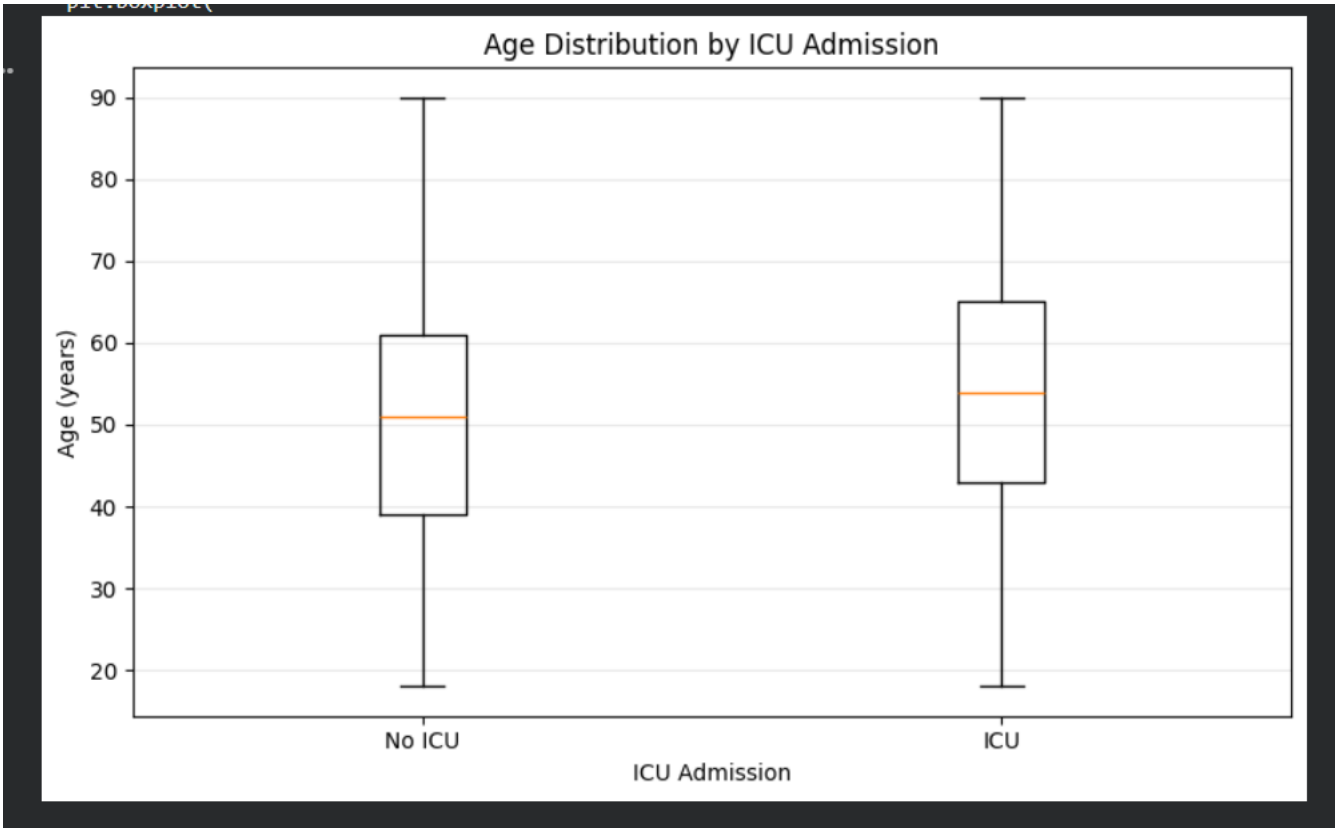
Age by ICU admission

Radiomics correlation matrix



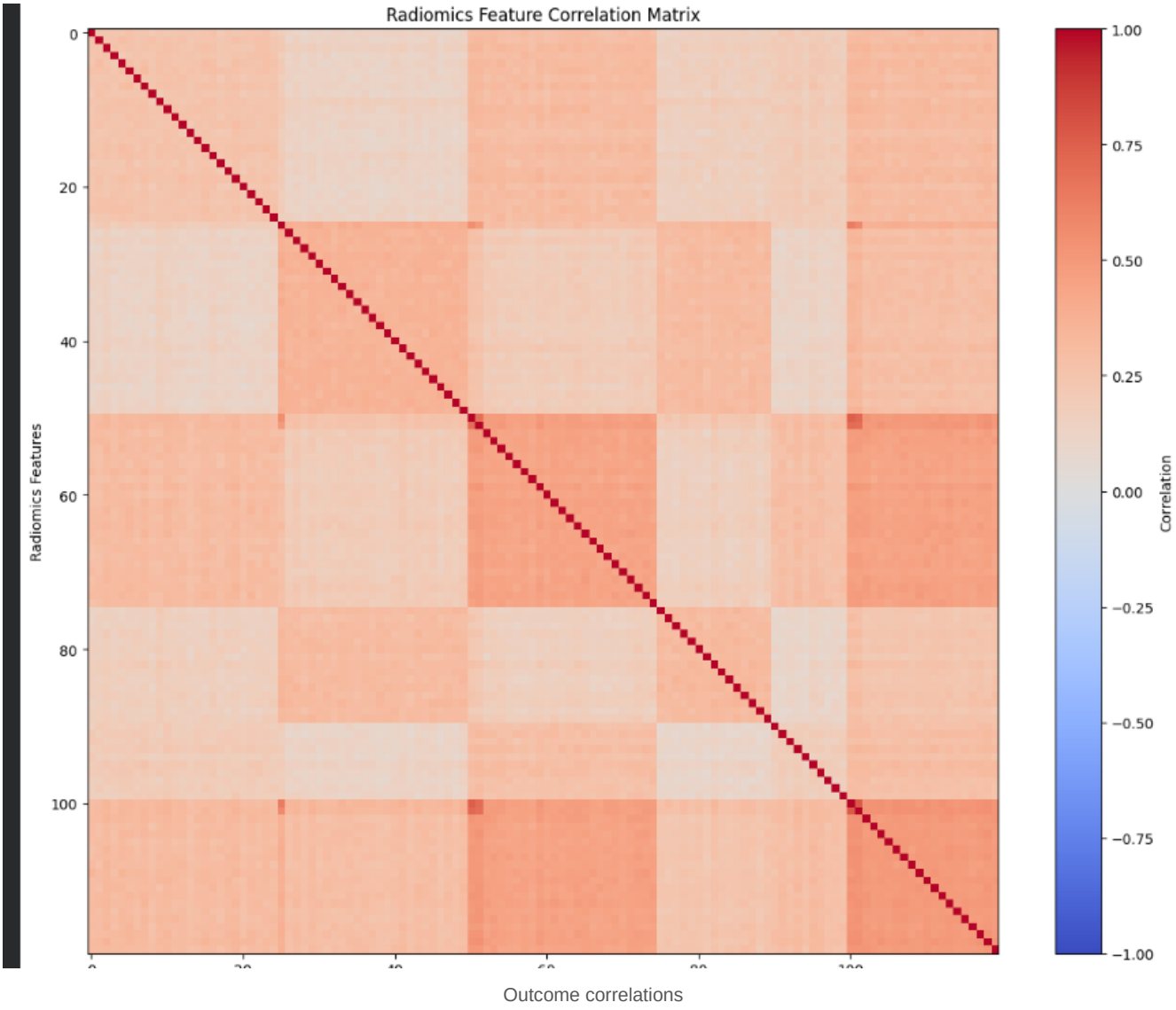
Radiomics correlation matrix

Clinical and laboratory correlations

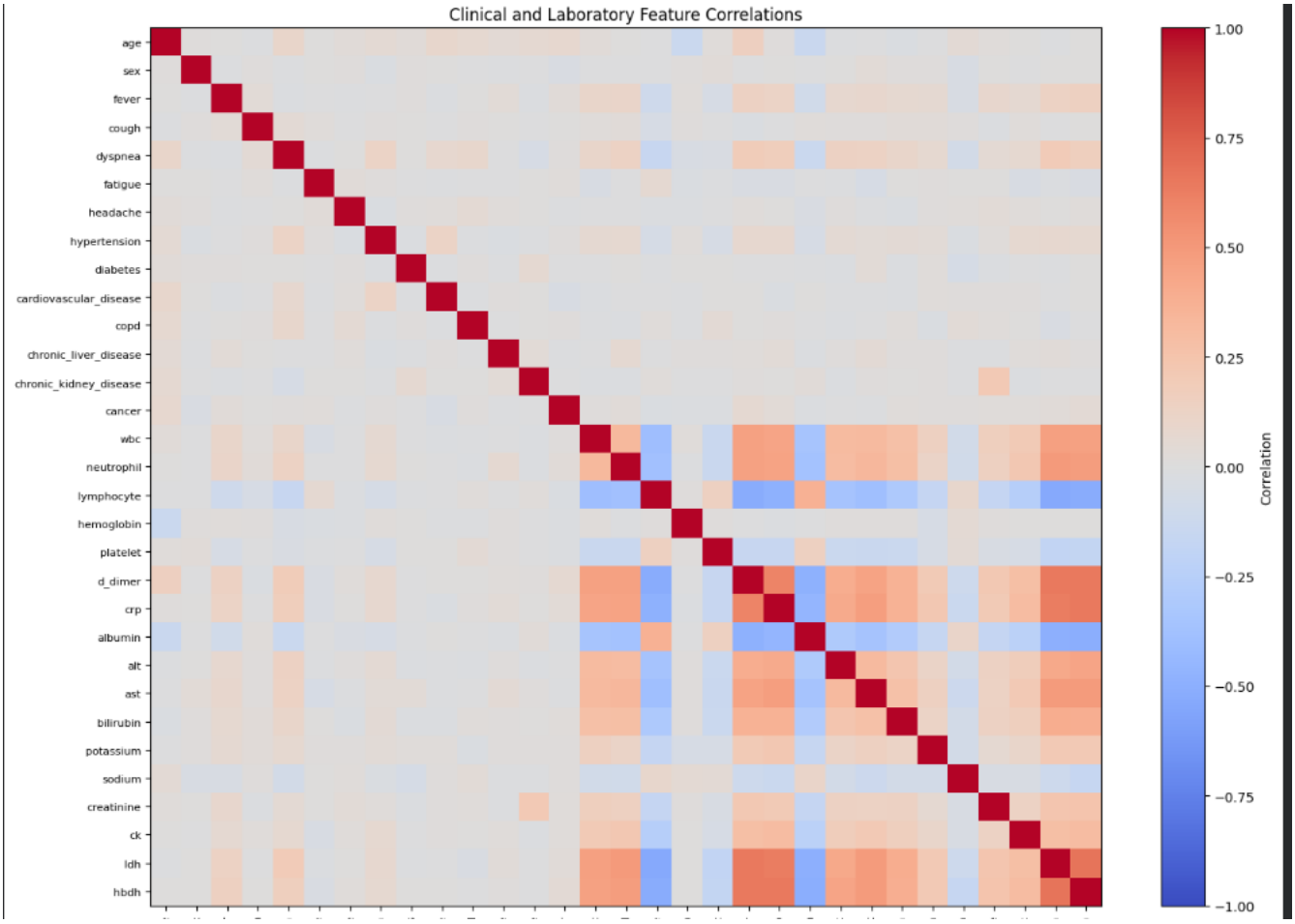


Clinical and laboratory correlations

Outcome correlations

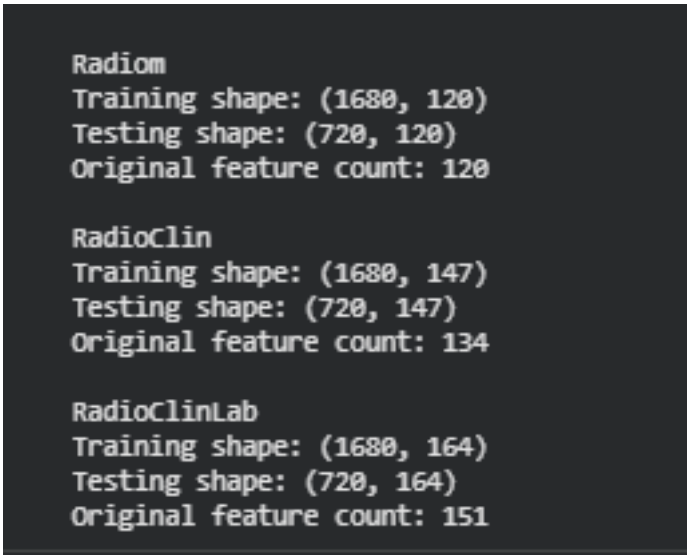


Missing-data percentage plot



Missing-data percentage plot

Feature-group matrix dimensions



Feature-group matrix dimensions

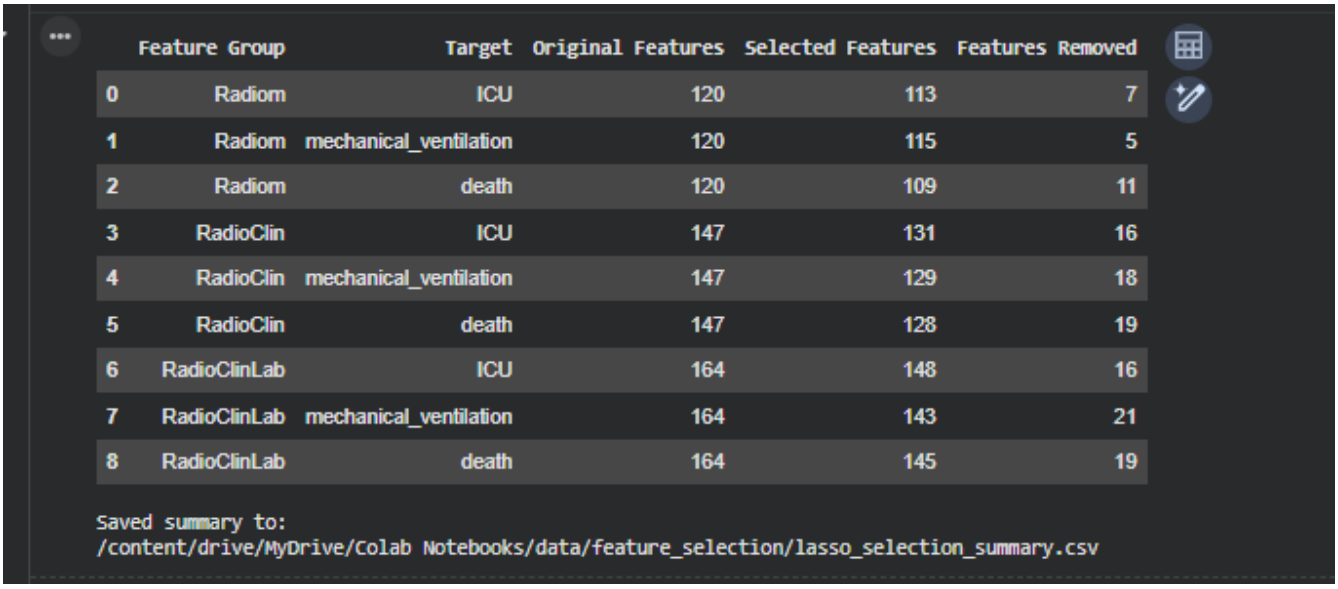
LASSO selection summary

```
Radiom: train=(1680, 120), test=(720, 120), features=120
RadioClin: train=(1680, 147), test=(720, 147), features=147
RadioClinLab: train=(1680, 164), test=(720, 164), features=164

Data verification passed.
```

LASSO selection summary

LASSO coefficient examples



	Feature Group	Target	Original Features	Selected Features	Features Removed
0	Radiom	ICU	120	113	7
1	Radiom	mechanical_ventilation	120	115	5
2	Radiom	death	120	109	11
3	RadioClin	ICU	147	131	16
4	RadioClin	mechanical_ventilation	147	129	18
5	RadioClin	death	147	128	19
6	RadioClinLab	ICU	164	148	16
7	RadioClinLab	mechanical_ventilation	164	143	21
8	RadioClinLab	death	164	145	19

Saved summary to:
/content/drive/MyDrive/Colab Notebooks/data/feature_selection/lasso_selection_summary.csv

LASSO coefficient examples

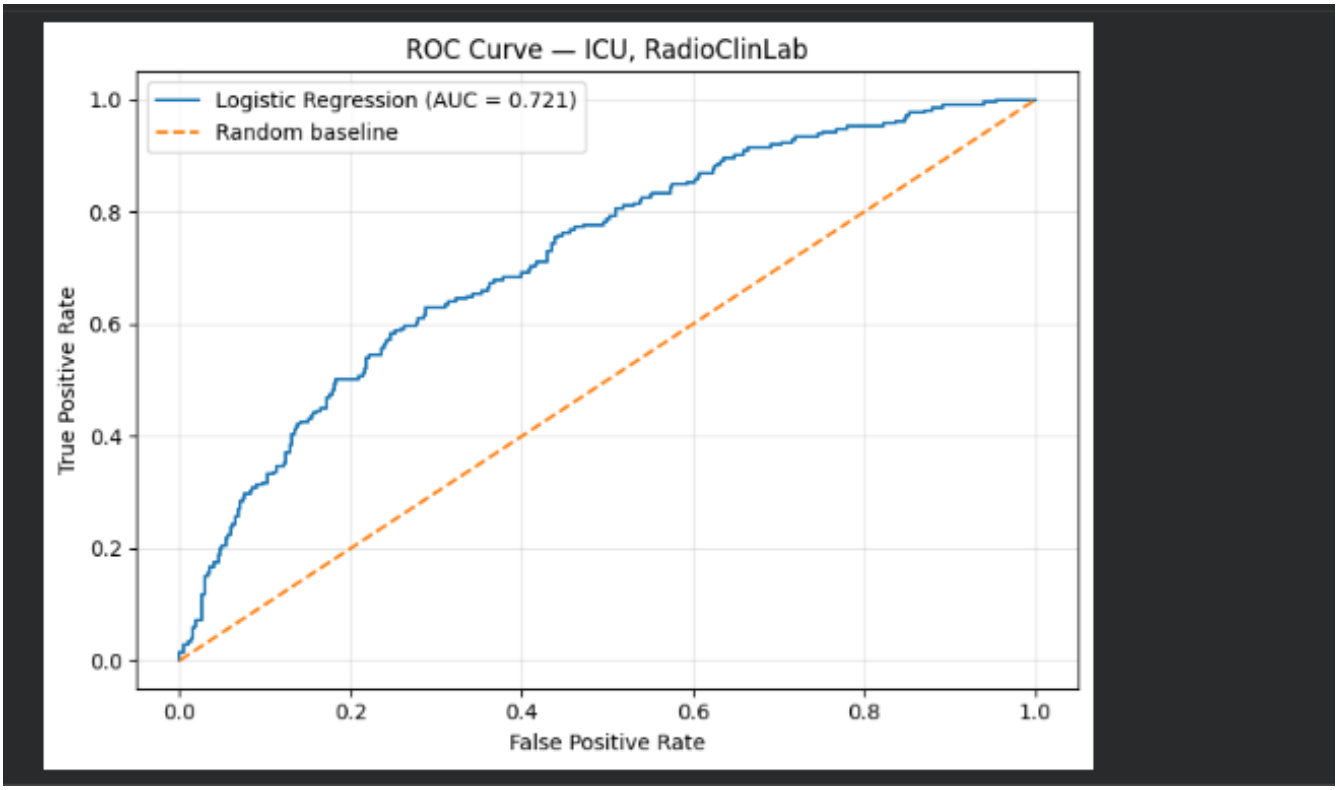
Model comparison excerpt

All model experiments completed.

	Feature Group	Target	Model	Selected Features	Train Rows After SMOTEENN	Accuracy	Precision	Recall	F1 Score	ROC-AUC	PR-AUC (Average Precision)
0	Radiom	ICU	Logistic Regression	113	1170	0.305556	0.296192	0.995261	0.456522	0.716887	0.502241
1	Radiom	ICU	Random Forest	113	1170	0.294444	0.293463	1.000000	0.453763	0.630620	0.368215
2	Radiom	ICU	SVM	113	1170	0.302778	0.295933	1.000000	0.456710	0.632231	0.369943
3	Radiom	ICU	MLP	113	1170	0.298611	0.294693	1.000000	0.455232	0.615136	0.423261
4	Radiom	ICU	LightGBM	113	1170	0.305556	0.296765	1.000000	0.457701	0.644215	0.377242
5	Radiom	mechanical_ventilation	Logistic Regression	115	1218	0.315278	0.300429	0.981308	0.460022	0.695098	0.512505
6	Radiom	mechanical_ventilation	Random Forest	115	1218	0.313889	0.301700	0.995327	0.463043	0.665403	0.407005
7	Radiom	mechanical_ventilation	SVM	115	1218	0.326389	0.304473	0.985981	0.465270	0.569235	0.336600
8	Radiom	mechanical_ventilation	MLP	115	1218	0.316667	0.302557	0.995327	0.464052	0.694812	0.519741
9	Radiom	mechanical_ventilation	LightGBM	115	1218	0.379167	0.321045	0.976636	0.483237	0.629276	0.355969

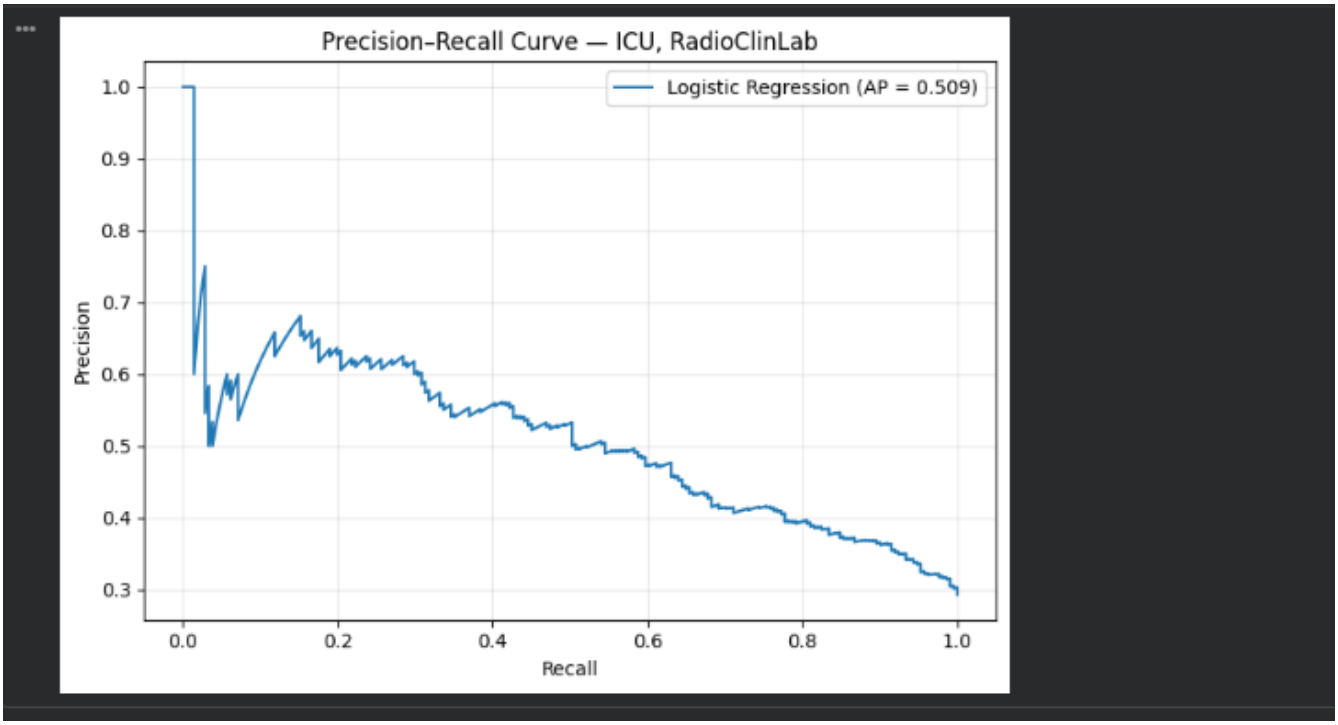
Model comparison excerpt

ICU ROC curve



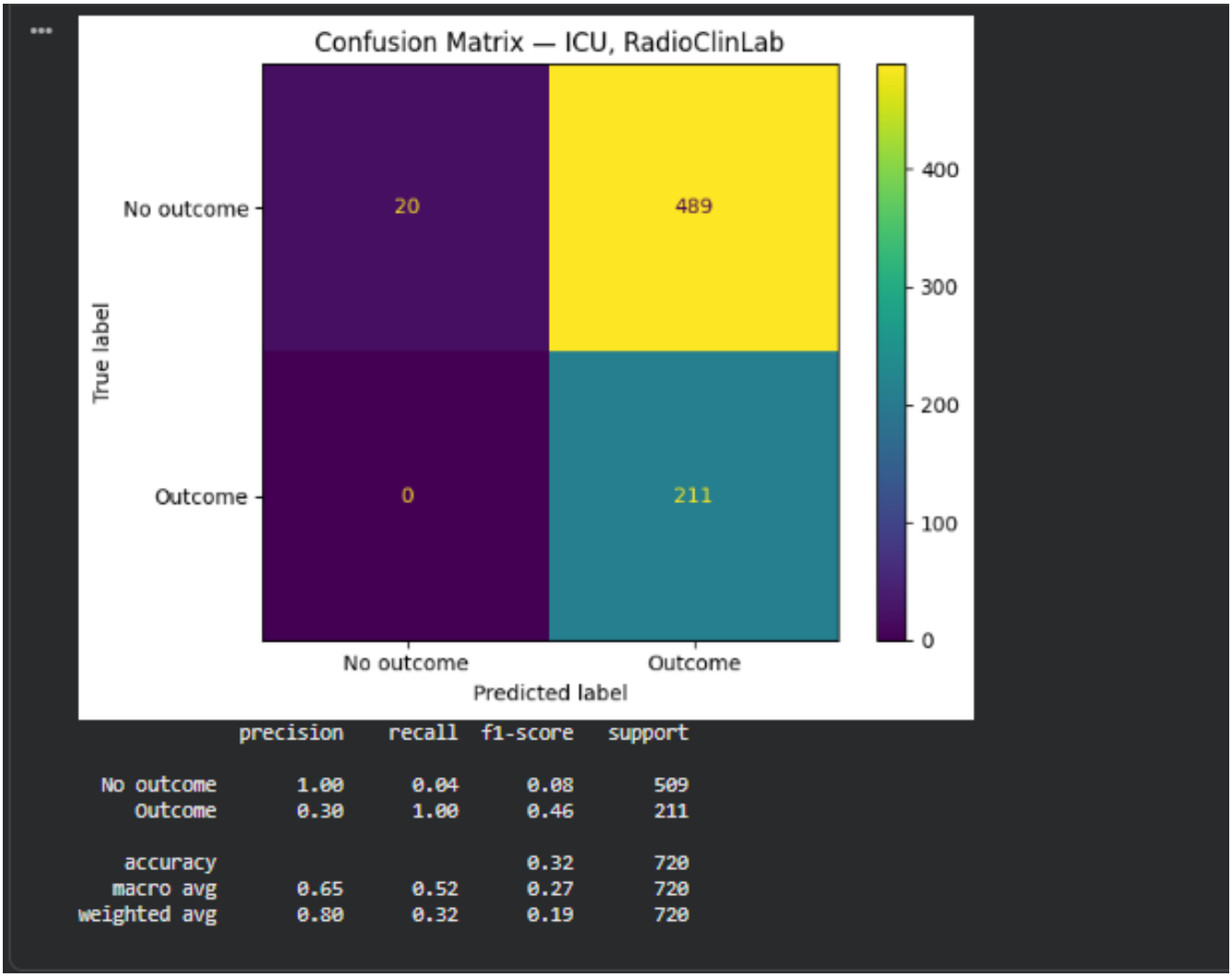
ICU ROC curve

ICU precision-recall curve



ICU precision-recall curve

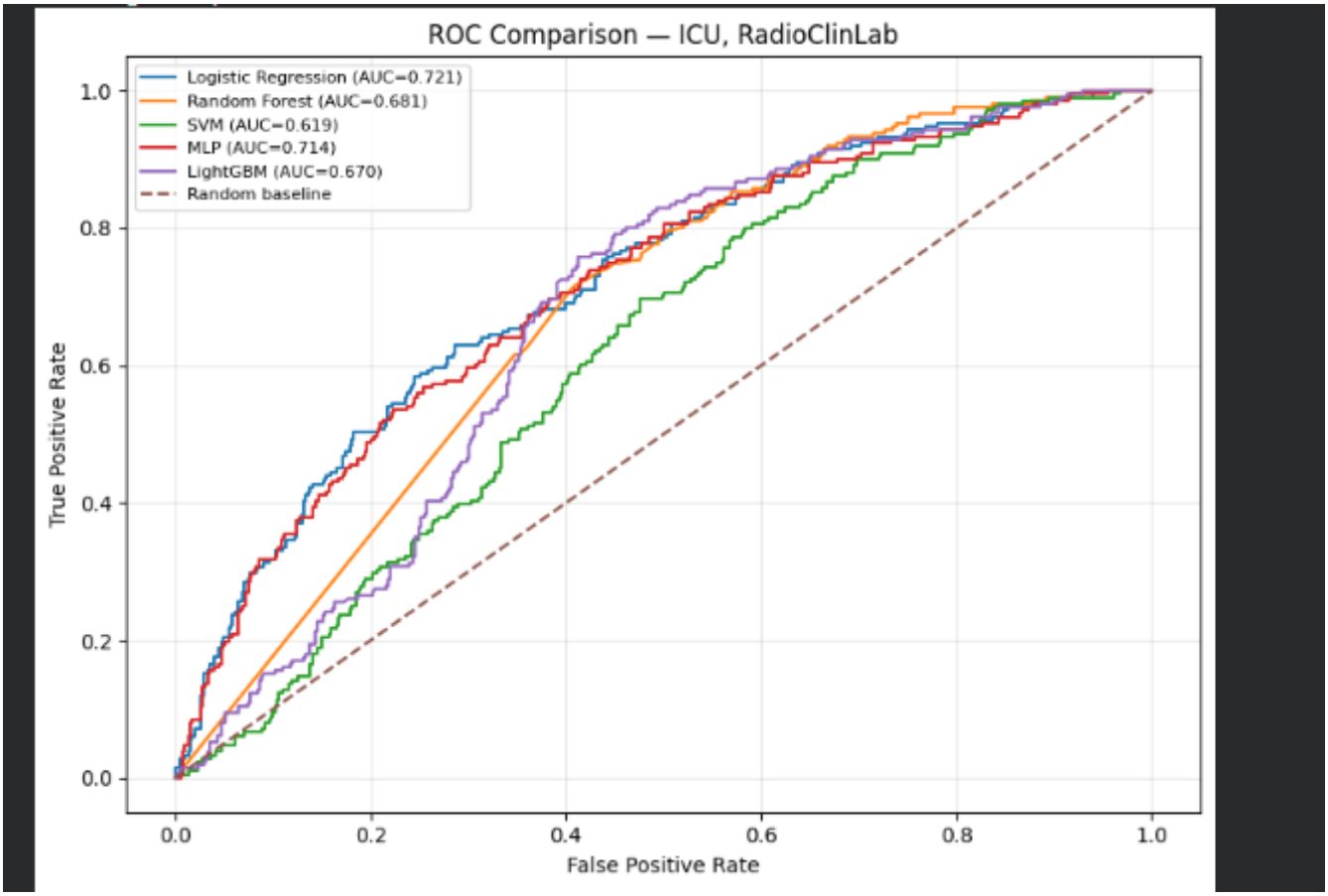
ICU confusion matrix and classification report



	precision	recall	f1-score	support
No outcome	1.00	0.04	0.08	509
Outcome	0.30	1.00	0.46	211
accuracy			0.32	720
macro avg	0.65	0.52	0.27	720
weighted avg	0.80	0.32	0.19	720

ICU confusion matrix and classification report

ROC comparison across ICU models



ROC comparison across ICU models

Best model summary

	Target	Feature Group	Model	ROC-AUC	F1 Score
0	ICU	RadioClinLab	Logistic Regression	0.721431	0.463227
1	death	RadioClinLab	Random Forest	0.729500	0.294118
2	mechanical_ventilation	RadioClinLab	Logistic Regression	0.736582	0.466742

Saved best-model summary.

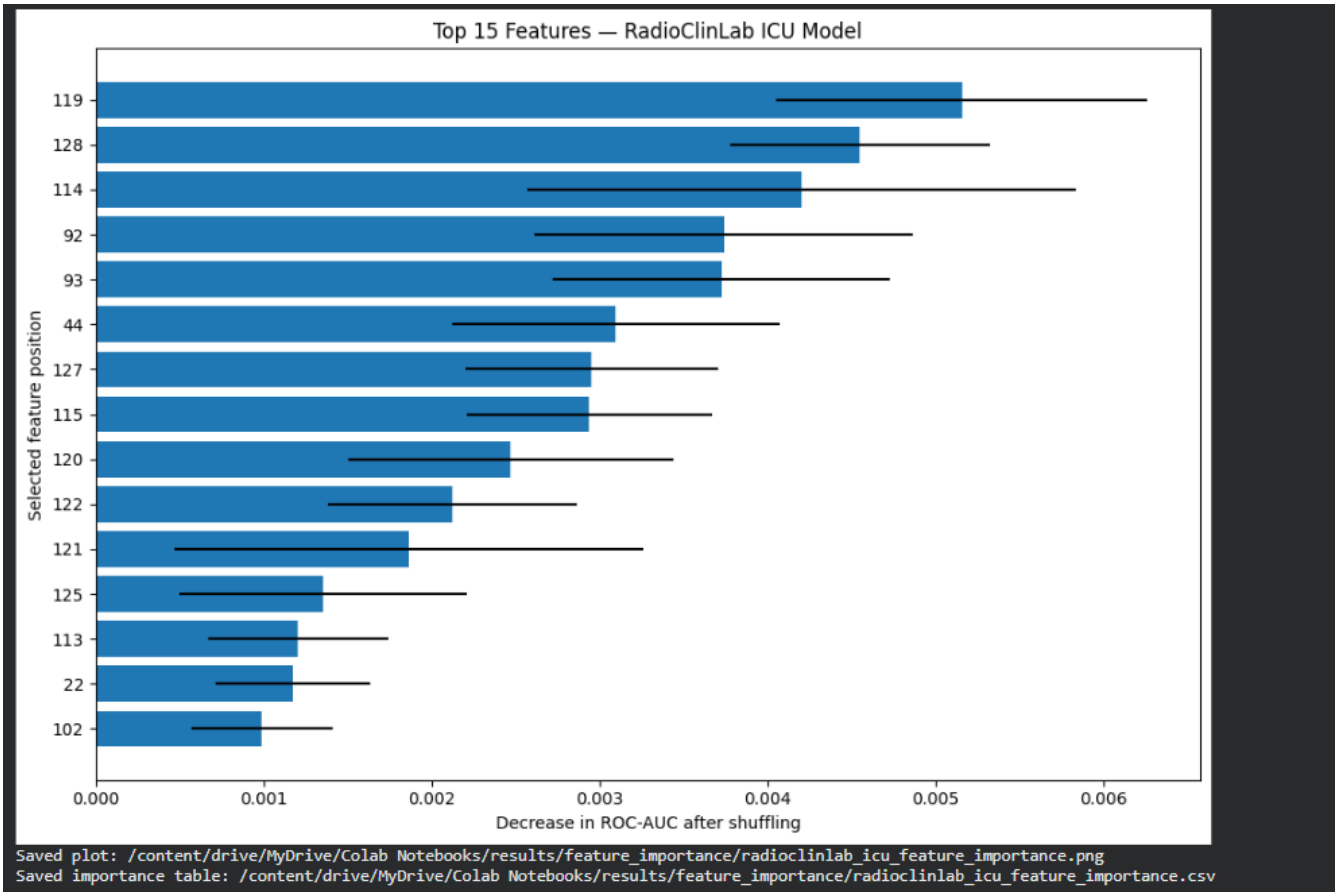
Best model summary

Feature-selection summary screenshot

	Feature Group	Target	Original Features	Selected Features	Features Removed
0	Radiom	ICU	120	113	7
1	Radiom	mechanical_ventilation	120	115	5
2	Radiom	death	120	109	11
3	RadioClin	ICU	147	131	16
4	RadioClin	mechanical_ventilation	147	129	18

Feature-selection summary screenshot

Permutation feature importance plot



Permutation feature importance plot